APEX RETAIL GROUP | PRACTICAL CLASS TASKS

DOCUMENT: CLASS TASKS (SESSION 2)

SCENARIO: FULFILLMENT SLA AUDIT & DATA VALIDATION

INSTRUCTIONS FOR LEARNER: Complete the following four tasks in your Jupyter Notebook during class. Follow the structured workflow: **Business Rule Mapping -> Code Execution -> Data Validation -> Business Interpretation.** Do not peek at instructor solution keys until completed.

TASK 1: BUSINESS LOGIC MAPPING

Marcus Vance (Operations Lead) wants to categorize fulfillment processing times into three operational risk bands:

- "Low Risk": Processing time is less than or equal to 10.0 hours.
- "Medium Risk": Processing time is greater than 10.0 hours but less than or equal to 20.0 hours.
- "High Risk": Processing time is greater than 20.0 hours.

1. Write out the conditional logic structure using pseudo-code or plain language (`if`, `elif`, `else`).
2. Identify what logic error occurs if an analyst checks `hours > 10.0` BEFORE checking `hours > 20.0` in a standard top-down conditional statement.

TASK 2: WRITING A REUSABLE EVALUATION FUNCTION

In your Jupyter Notebook, write a Python function named `classify_risk(processing_hours)`:

1. The function must accept one parameter: `processing_hours`.
2. Inside the function, use `try` / `except` to attempt converting `processing_hours` to a `float`. If conversion fails, return string `"Invalid Data"`.
3. Apply the risk criteria from Task 1:
    - Return `"Low Risk"` if hours <= 10.0
    - Return `"Medium Risk"` if hours <= 20.0
    - Return `"High Risk"` if hours > 20.0
4. Test your function with four test calls: `classify_risk(8.5)`, `classify_risk(15.0)`, `classify_risk(25.0)`, and `classify_risk("MISSING")`. Print the results.

TASK 3: BATCH PROCESSING WITH A FOR LOOP

Marcus provides a list of logged processing times from Distribution Center Alpha:

```
dc_alpha_times = [12.0, 8.5, "CORRUPTED", 22.5, 19.0, 31.0, 9.0, 14.5,
"UNKNOWN", 27.0]
```

Write Python code to:

1. Initialize counters for `low_risk_count`, `medium_risk_count`, `high_risk_count`, and `invalid_count`.
2. Use a `for` loop to iterate through `dc_alpha_times`.
3. Inside the loop, pass each item to your `classify_risk()` function.
4. Update the respective counters based on the returned risk category.
5. Print the total count for each risk category and the total number of valid orders processed.

TASK 4: METRIC CALCULATION & EXECUTIVE MEMO

Review your counter totals from Task 3:

1. Calculate the percentage of valid orders classified as "High Risk". Formula: (High Risk Count / Total Valid Orders) * 100
2. Write a 2-3 sentence operational recommendation to Marcus Vance summarizing the operational health of Distribution Center Alpha and calling out any data logging concerns.